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Systematic Review: Wearable Technology for Cardiac Rhythm Monitoring

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Abstract: Wearable technology for cardiac rhythm monitoring has rapidly emerged as a non-invasive, cost-effective, and scalable solution for the early detection, diagnosis, and management of arrhythmias. This systematic review aims to synthesize current literature on the clinical accuracy, usability, and limitations of wearable devices for cardiac rhythm assessment. A comprehensive search was conducted across PubMed, Scopus, and IEEE Xplore databases covering publications from January 2015 to April 2025. Inclusion criteria focused on studies evaluating smartwatches, chest straps, and wearable patches with electrocardiographic (ECG) or photoplethysmography (PPG) capabilities. Results demonstrate a growing body of evidence supporting the feasibility and accuracy of wearables in identifying atrial fibrillation, premature contractions, and heart rate variability. Devices such as the Apple Watch, Fitbit, and AliveCor KardiaMobile have shown promising sensitivity and specificity when compared to clinical gold standards. However, challenges remain in terms of motion artifacts, signal noise, regulatory approvals, and integration with clinical workflows. This review underscores the potential of wearable cardiac monitors in preventive cardiology and telehealth, while highlighting the need for standardization and long-term validation in diverse populations.

Keywords: wearable ECG, photoplethysmography, atrial fibrillation, cardiac arrhythmia, remote monitoring, heart rate variability, telecardiology

INTRODUCTION

Cardiovascular diseases (CVDs) are the leading cause of death globally, with atrial fibrillation (AF) alone contributing significantly to increased risks of stroke, heart failure, and mortality [1]. Traditional cardiac monitoring methods like Holter monitors and implantable loop recorders are either limited in duration or invasive and expensive [2]. These limitations have created a demand for continuous, scalable, and non-invasive monitoring solutions that can be applied in everyday settings. In this context, wearable technology has emerged as a game-changing approach to cardiac rhythm assessment [3].

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The growing burden of cardiovascular diseases has necessitated a paradigm shift toward preventive and proactive healthcare strategies. This shift aligns with global health objectives to reduce premature mortality from non-communicable diseases [4]. Wearable devices— ranging from wristbands to adhesive patches and chest straps—are now capable of monitoring heart rate, rhythm irregularities, and variability patterns in real-time, offering unprecedented opportunities for early detection and intervention [5].

The devices use ECG sensors or photoplethysmography (PPG) technology. ECG sensors read the electrical activity of the heart – PPG sensors read blood volume changes with an optical technology. ECG-based wearables, including the AliveCor KardiaMobile and Apple Watch Series 4 and above, have demonstrated sensitivity and specificity similar to conventional approaches for AF detection [6,7]. PPG-based devices on the other hand can offer less precise measurements, but continuous passive monitoring and better wearability for long-term use [8].

It is not just the portability but the longitudinal nature of the physiological data that separates wearable devices from clinical settings. Such information can provide insight into heart rate variability (HRV), circadian rhythms, and arrhythmic burden in a cost-effective way [9]. These devices have also enabled patients to play a role in their own health, a practice that is known to enhance adherence and results [10].

But as the explosion of commercial wearables continues, so too does the need for data accuracy, compatibility with the healthcare system, and regulatory scrutiny. Sensors used in the devices substantially vary in terms of quality, algorithm employed, and capability to interpret signals [11]. Motion artifact, variability of skin tone and user compliance are still challenges facing the reliability of data collected [12]. For example, darker skin colour and tattoos can interfere with PPG obtained by light, and poor attachment of devices (and dry skin) can result in signal artifacts [13].

These shortcomings aside, wearables are more popular than ever. According to new industry reports, the global consumer health wearables market is projected to reach over $ 100 billion by 2027, based on an increased awareness and adoption of devices, with cardiac monitoring applications forming a significant share [14]. Artificial intelligence (AI) and machine learning advancements have further improved wearables' diagnostic capabilities.. Algorithms empowered with AI technologies can consider the complex ECG or PPG data and accordingly detect even arrhythmias with remarkable accuracy [15]. These updates have attracted some regulatory attention; for instance, the FDA has granted certain wearable ECG capabilities for clinical use [16].

Another factor fueling wearable adoption is the COVID-19 pandemic, which highlighted the importance of remote patient monitoring to reduce hospital visits and exposure risks [17]. Patients with known cardiovascular conditions, particularly those at risk of AF, benefitted from wearable ECG and PPG devices that allowed real-time symptom tracking and event-triggered recordings [18]. Clinicians were able to receive automated alerts for irregular rhythms, facilitating timely interventions without in-person appointments [19].

Apart from AF, the research on wearables as arrhythmia detectors is expanding to include extra types of arrhythmias like bradycardia, tachycardia and premature ventricular contractions (PVCs) [20]. Some other research works have also investigated their background on assessing HRV as a biomarker for the central autonomic nervous system function, which is applicable to the heart failure, diabetes, and sleep apnea [21].

With the increasing capabilities and clinical utility of wearable ECG monitors, a synthesis of the literature was required to ascertain their efficacy, limitations, and future directions. This review seeks to examine the available evidence on the use of wearable technology for cardiac rhythm monitoring, including their diagnostic accuracy, ease of use, patient compliance, and linked data with telemonitoring systems. The most frequently used devices and technologies are reviewed, their status of approval, and patient/clinician implications are explored.

Key Advantages of Wearable Technologies for Cardiac Rhythm Monitoring

Advantage	Description
Continuous Monitoring	Enables long-term and real-time tracking of cardiac rhythms
Early Detection	Facilitates identification of arrhythmias and cardiac events at an early stage
Convenience	Provides a non-invasive, comfortable, and user-friendly monitoring option
Integration with Health Tools	Allows connectivity with smartphones and health applications for data analysis

MATERIALS AND METHODS

A systematic review protocol was developed

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following PRISMA 2020 guidelines [22]. A comprehensive search strategy was applied to PubMed, Scopus, and IEEE Xplore databases, focusing on peer-reviewed literature published between January 2015 and April 2025. Keywords used in the search included "wearable cardiac monitor," "ECG smartwatch," "photoplethysmography arrhythmia," "atrial fibrillation detection wearable," and "remote heart monitoring."

Inclusion and Exclusion Criteria To ensure the relevance and scientific rigor of included studies, the following inclusion criteria were applied:

- Original peer-reviewed studies or systematic reviews.
- Research involving wearable devices equipped with ECG or PPG technology.
- Studies reporting on diagnostic accuracy, usability, or clinical outcomes.
- Human studies including clinical trials, cohort studies, and cross-sectional analyses.

Exclusion criteria were:

- Non-English language articles.
- Editorials, letters, or conference abstracts without full-text availability.
- Studies focusing solely on step counts or fitness without rhythm monitoring capability.

Data Extraction and Analysis Two independent reviewers screened titles and abstracts, followed by full-text review. Discrepancies were resolved through consensus or a third reviewer. Extracted data included:

- Author and year
- Study population
- Type of wearable device (ECG/PPG)
- Arrhythmia types monitored
- Diagnostic performance metrics (sensitivity, specificity)
- Validation method (comparison with Holter, clinical ECG, etc.)

Quality assessment was conducted using the QUADAS-2 tool for diagnostic accuracy studies [23]. Statistical heterogeneity between studies was acknowledged but not pooled due to methodological diversity.

PRISMA Flow Chart

The PRISMA diagram below summarizes the study selection process:

Stage	Number of Records
Records identified through database searching	326
Records after duplicates removed	291
Records screened	291
Full-text articles assessed	78
Studies included in review	32

This flow chart reflects a structured selection pipeline in accordance with PRISMA principles [24,25]. Although a large number of initial records were identified, stringent criteria and methodological consistency led to a focused and high-quality final sample.

The methodology applied ensures reproducibility, transparency, and minimal bias in selecting studies relevant to the evaluation of wearable technologies for cardiac rhythm monitoring. The reliance on peer-reviewed and clinically validated data supports the robustness of this review's conclusions [26].

RESULTS

The results of this systematic review are derived from 32 studies meeting inclusion criteria, spanning randomized controlled trials, prospective cohort studies, and validation reports of wearable ECG and PPG devices. Analysis reveals promising diagnostic performance for atrial fibrillation (AF) and heart rate monitoring, especially in ambulatory environments.

The ECG-based wearables Devices (e.g., the Apple Watch [Series 4 and above], AliveCor KardiaMobile, and Withings ScanWatch) showed high sensitivity and specificity for detection of AF in many clinical validation studies. The former: The sensitivity and specificity were between 93%–98% and 90%–97%, respectively, when compared with 12-lead ECG or Holter monitors [27]. For example, single-lead ECG technology in Apple Watch was highly accurate in detecting rhythm abnormalities in hospitalized and community-based settings.

Photoplethysmography (PPG) Devices PPG sensors, which were included in devices such as Fitbit Sense, Garmin Vivosmart, and Samsung Galaxy Watc,h were tested predominantly for continuous HR monitoring and arrhythmia detections. While slightly less accurate than ECGs in general, certain papers had sensitivities higher than 85% for

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AF detection and greater than 95% for rest state HR accuracy [28]. Despite that, motion artifacts and skin tone variation remained unsolved, particularly during movement or low perfusion.

Arrhythmias Other than Atrial Fibrillation. Only a limited number of studies addressed ventricular arrhythmias and bradycardia detection. Some patch-like devices (e.g., Zio Patch and Biobeat) showed the ability to detect premature ventricular contractions (PVCs), supraventricular tachycardia, and pauses over three seconds. However, the false positives were a problem in high-movement settings [29]. Further statistical algorithms and real-time clinician data validation are needed to extend clinical utility beyond AF.

Usability and Patient Adherence. User acceptance and adherence were strong, particularly for smartwatch-based devices. The studies observed daily use of greater than 80%, and the device comfort and app usability were reported as positive [30]. Older users took longer to onboard and dropped out at a slightly higher rate, typically, because of complicated interfaces or syncing problems. On the other hand, a younger patient population interacted more frequently with their health data and remote alerts.

Data Transmission and Clinical Integration A significant barrier remains the interoperability between consumer-grade devices and electronic health records (EHRs). While platforms like Apple HealthKit and KardiaPro offer clinician dashboards, full integration with hospital systems was rarely implemented in the reviewed studies [31]. Intermittent data gaps, transmission latency, and limited clinician feedback loops restricted the clinical decision-making value in real-time.

Summary Table of Diagnostic Performance

Device	Sensor Type	Primary Rhythm Detection	Sensitivity (%)	Specificity (%)	FDA/CE Approved
Apple Watch Series 6	ECG	Atrial Fibrillation	97	94	Yes
KardiaMobile 6L	ECG	AF, Bradycardia	98	95	Yes
Fitbit Sense	PPG	Heart Rate, AF (limited)	87	91	Partial
Samsung Galaxy Watch	PPG	AF (algorithm-based)	85	89	No
Biobeat Chest Patch	ECG	PVCs, HR Variability	92	90	Yes

Limitations Observed in Studies Heterogeneity in study design, participant demographics, and outcome measures posed challenges in cross-comparison. Many studies excluded high- risk or multi-comorbidity groups, leading to a limited generalizability of findings [32]. Moreover, few trials exceeded 6-month follow-ups, raising concerns about long-term reliability and sustained adherence.

The results support growing confidence in wearable devices for remote cardiac rhythm monitoring, particularly in AF detection. However, refinement in algorithms, better user education, and improved system integration are needed to realize their full clinical potential.

DISCUSSION

Wearable technology has revolutionized how cardiac rhythm disturbances are detected and managed, bridging the gap between patient autonomy and clinical oversight. As evidenced in this review, wearable ECG and PPG-based monitors have demonstrated promising diagnostic capabilities, particularly for atrial fibrillation (AF), and offer an accessible means of rhythm monitoring outside traditional healthcare settings [33]. This paradigm shift aligns with broader healthcare trends toward decentralization and patient-centered care, wherein data can be collected continuously, interpreted remotely, and used for early intervention [34].

Technological advancements in sensors, longer battery life, and the incorporation of artificial intelligence AI into portable platforms have played an important role in the adoption of wearable cardiac monitors. Real-time AI-enabled algorithms now help in detecting arrhythmias and enhance the predictive capability of these devices while decreasing physician workload [35]. For example, it has been shown that deep learning networks have the ability to detect complex rhythms such as PVCs and bigeminy with a

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sensitivity that matches expert viewers as long as the input signal quality is sufficiently high.

However, limitations remain due to accuracy inconsistency between skin colours and physiological state. It has been indicated that PPG based-devices can present with unreliable performance in dark skin, because of low signal penetration that can cause inequalities in detecting arrhythmias [36]. Moreover, variability of heart rate may arise from temperature, fluid (hydration levels) or stress status, which can introduce artifacts and risk diagnostic accuracy [37].

Lack of EHR integration is another major obstacle that prevents broad adoption of wearable cardiac devices in clinical practice. A lot of consumer platforms are siloed ecosystems, and are not able to release data back and forth to both the patient and the caregiver [38]. Partial integration is possible through platforms such as Apple HealthKit and Fitbit SDK but they necessitate application of middleware or manual input into clinical workflows.

The ethical and privacy implications of wearable cardiac monitors also merit consideration. The continuous transmission of physiological data raises concerns about data ownership, consent, and cybersecurity [39]. Breaches in health data can have profound implications for patient trust and regulatory compliance, particularly under frameworks like GDPR in Europe and HIPAA in the United States. Transparent data governance frameworks are essential to maintaining user confidence and fostering responsible innovation.

Long-term adherence to wearable monitoring is another area requiring attention. While short- term engagement is high, studies report a gradual decline in consistent use over 3 to 6 months, especially in older adults [40]. Factors influencing adherence include device comfort, app usability, user education, and the perceived clinical value of monitoring. Strategies to enhance sustained use include personalized feedback, behavioral nudges, and integration with telehealth coaching services.

Economically, wearable monitors show promise in reducing healthcare costs associated with delayed arrhythmia diagnosis and unnecessary emergency room visits. However, reimbursement models remain unclear in many healthcare systems. Payers and policymakers need to establish evidence-based reimbursement pathways for remote cardiac monitoring to encourage broader adoption [41]. This includes recognizing wearable diagnostics in insurance billing codes and value-based care frameworks.

Importantly, the COVID-19 pandemic has fast-forwarded the adoption of remote patient monitoring, and wearables have become essential to those patients with limited access to in- person cardiac care [42]. Wearables facilitated such continuity of care, but infection risk, and multiple health systems adopted them to triage and manage highly susceptible populations during times of peak pandemic.

Future research should seek to eliminate the reported disparities in demographic representation, particularly for the underserved populations. The majority of the current work is dominated by high income countries and urban areas, and there is a lack of information regarding the effectiveness of wearables in rural and low-resourced environments [43]. Furthermore, children and teenagers are underrepresented in wearables deployment, although usage by younger purchasers is increasing.

And lastly wearable cardiac monitoring is going to be one of the main players in preventive cardiology and population-based therapy. These technologies allow the identification of at- risk individuals prior to the onset of symptoms, allowing for earlier lifestyle modifications, treatment adjustments and specialist referrals, which could in turn prevent disastrous CVD events [44]. Translation of technological capacity into scalable health solutions requires partnership between device makers, clinical research, and public health.

The future of wearable cardiac monitoring lies in the convergence of biosensor innovation, AI- driven analytics, and integrative digital platforms. For wearable devices to reach their full potential, efforts must be directed toward improving sensor accuracy, enhancing user experience, and establishing robust clinical validation in diverse real-world populations. Furthermore, global regulatory harmonization and evidence-based guidelines are necessary to standardize evaluation protocols, facilitate clinical adoption, and safeguard patient safety [45].

CONCLUSION

Wearable technology for cardiac rhythm monitoring represents a transformative advancement in the delivery of cardiovascular care, offering unprecedented opportunities for real-time, continuous, and non-invasive assessment of arrhythmias such as atrial fibrillation. The evidence gathered in this review underscores the clinical validity and growing acceptance of ECG- and PPG-based wearable devices in both diagnostic and preventive cardiology. While promising results have been demonstrated in terms of sensitivity, specificity, and patient engagement, widespread implementation continues to face hurdles including device accuracy under varying conditions, user

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adherence, data integration with healthcare systems, and regulatory clarity. Moving forward, it is imperative to strengthen interdisciplinary collaboration, ensure equitable access, and foster robust longitudinal studies that assess not just diagnostic efficacy but also long-term health outcomes. If these challenges are effectively addressed, wearable cardiac monitors have the potential to significantly reduce the global burden of cardiovascular disease through earlier detection, improved monitoring, and more personalized interventions.

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